








# Modelling Financial Markets

## Seminar 9: Realised Volatility — Hands-on




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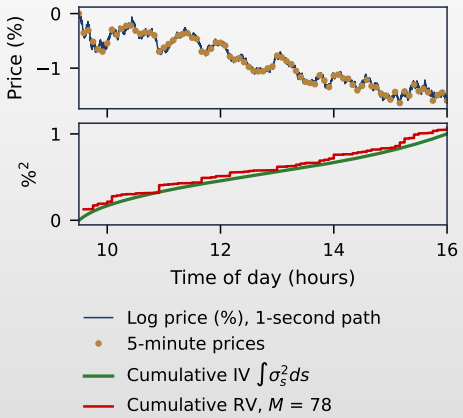
# Does Intraday Measurement Improve a Volatility Forecast?

- Question of the seminar
  - **how precisely does one day of intraday prices measure volatility, and does the measurement improve a forecast?**
- Two parts, in this order
  - Part I, measurement and diagnosis: A1, A2, B1, B2, A3
  - Part II, models and forecasts: A4, B3, A5, A6, C1
- Deliverables
  - the answers to the A exercises
  - one figure and one table per B exercise
  - the C1 correction table
  - one interpretation sentence per exercise
- Notebook for today: open the seminar notebook in Colab; each task names its notebook section
- Background: Chapter 9 and Franke, Härdle & Hafner (2019)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro; Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

## Which Market Clock and Measurement Units?

- SPY is an Exchange-Traded Fund (ETF) tracking the S&P 500
- Bitcoin prices are in United States dollars (USD)
- Timestamps are in Coordinated Universal Time (UTC); SPY bars are converted to New York time
- Files in data/market/intraday: `SPY.US_5m.csv` and `BTC-USD.CC_5m.csv`, with `datetime_utc`
- A bar timestamp is the start of its interval
- Intraday chart axes show interval ends
- Annualisation uses 252 days for SPY and 365 for Bitcoin
- Returns are logarithmic, in percent; their squares are in percent squared
- C1 checks an artificial intelligence (AI) answer: for every number, state the asset, interval, unit and forecast origin

# Realised Variance




- Details: Chapter 9
- **Integrated variance** (IV): the variance the day actually delivered

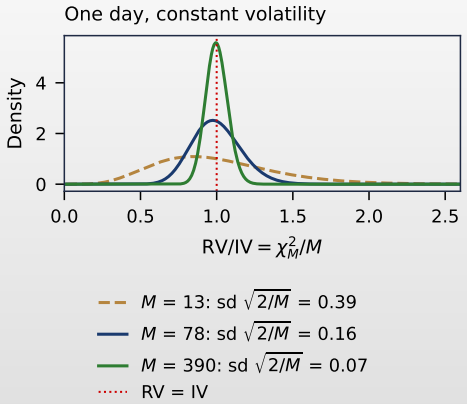
$$\mathrm{IV}_t = \int_{t-1}^{t} \sigma_s^2 \, ds$$

  - $\sigma_s$: spot (instantaneous) volatility at time $s$ inside day $t$
- **Realised variance** (RV) Andersen et al. (2001): the sum of the $M$ squared intraday returns

$$\mathrm{RV}_t = \sum_{i=1}^{M} r_{t,i}^2 \xrightarrow{p} \mathrm{IV}_t \quad (M \to \infty)$$

  - $r_{t,i} = 100\,(\ln P_{t,i} - \ln P_{t,i-1})$: $i$-th return of day $t$, in %; SPY: $M = 78$ five-minute returns
  - valid for a continuous price without noise; annualised volatility $\sqrt{252\,\mathrm{RV}_t}$ (%)
  - chart: a simulated day with U-shaped intraday volatility; RV tracks IV step by step

# How Precise Is One Day's RV?

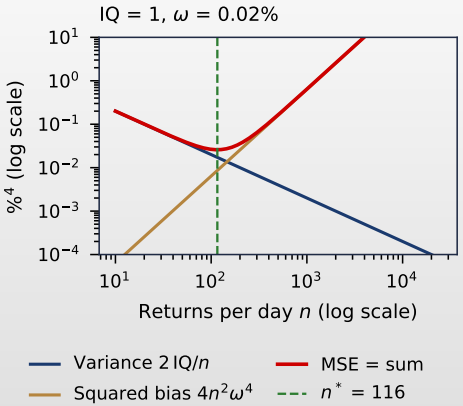



- Measurement error of RV for large $M$

$$\mathrm{RV}_t - \mathrm{IV}_t \;\approx\; N\Big(0,\; \frac{2\,\mathrm{IQ}_t}{M}\Big), \qquad \mathrm{IQ}_t = \int_{t-1}^{t} \sigma_s^4\, ds$$

  - $\mathrm{IQ}_t$: integrated quarticity; estimated by the realised quarticity $\mathrm{RQ}_t = \frac{M}{3}\sum_i r_{t,i}^4$
  - constant volatility: $\mathrm{RV}/\mathrm{IV} = \chi^2_M/M$, a chi-square with $M$ degrees of freedom divided by $M$, relative sd $\sqrt{2/M}$
- **Delta method**: for a smooth $g$, $\mathrm{se}(g(X)) \approx |g'(X)|\,\mathrm{se}(X)$
  - $g = \ln$: $\mathrm{se}(\ln \mathrm{RV}) \approx \mathrm{se}(\mathrm{RV})/\mathrm{RV}$; an interval built on the log scale and exponentiated stays positive
  - chart: the spread of $\mathrm{RV}/\mathrm{IV}$ for 13, 78 and 390 returns a day

# Microstructure Noise and the Sampling Frequency




- Observed log price = efficient price + noise

$$p_i^{\text{obs}} = p_i^* + \varepsilon_i, \qquad \varepsilon_i \text{ i.i.d. } (0, \omega^2)$$

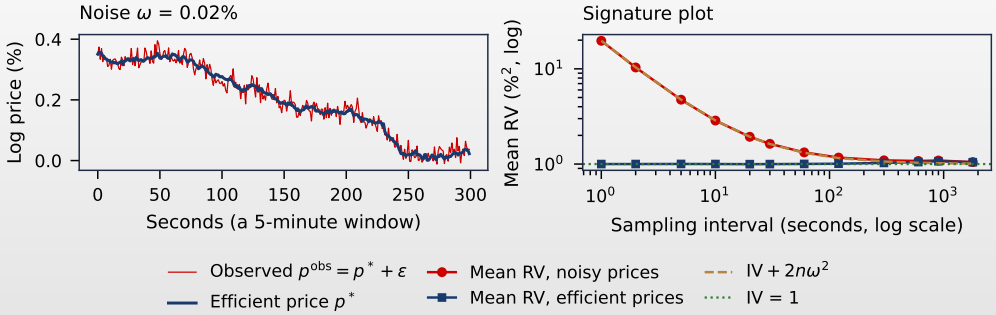
  - $\omega^2$: noise variance (bid–ask bounce, price ticks), $\varepsilon$ independent of $p^*$
  - each observed return $r_i = r_i^* + \varepsilon_i - \varepsilon_{i-1}$ carries two noise terms

- With $n$ returns: bias grows with $n$, sampling variance falls like $1/n$ Bandi & Russell (2008)

$$\mathbb{E}[\mathrm{RV}^{(n)}] = \mathrm{IV} + 2n\omega^2, \qquad \mathrm{MSE}(n) \approx \frac{2\,\mathrm{IQ}}{n} + 4n^2\omega^4$$

  - MSE (Mean Squared Error) = variance + squared bias; the minimising $n^*$ balances the two terms (chart: $\omega = 0.02\%$)
  - in practice 5 minutes is hard to beat Liu, Patton & Sheppard (2015)

# Noise in Prices and the Signature Plot




- Simulation: 150 days, $\text{IV} = 1\,\%^2$, noise $\omega = 0.02\%$
  - left: tick by tick the observed price oscillates around the efficient one
  - right, **signature plot**: mean RV against the sampling interval; noisy prices follow $\text{IV} + 2n\omega^2$, efficient prices stay at IV
  - a flat signature plot at the chosen interval means the noise bias is negligible there

# Sparse Grids, Averaging and the Block Bootstrap

- **Sparse grid** of $k \times 5$ minutes starting at offset $o \in \{0, \dots, k-1\}$

$$\mathrm{RV}_t^{(k,o)} = \sum_j \left(100\,\Delta \ln P\right)^{\mathbf{2}} \text{ on that grid}, \qquad \overline{\mathrm{RV}}_t^{(k)} = \frac{1}{k} \sum_{o=\mathbf{0}}^{k-\mathbf{1}} \mathrm{RV}_t^{(k,o)}$$

  - averaging over the $k$ shifted grids uses all prices and lowers the sampling variance, without more noise bias
  - ratio of means $\sum_t \mathrm{RV}_t^{(k)} / \sum_t \mathrm{RV}_t^{(\mathbf{5\ min})}$: near 1 when the noise bias at 5 minutes is negligible

- **Moving-block bootstrap** Künsch (1989): a confidence interval for a statistic of dependent data
  - draw random starting days, join blocks of $b$ consecutive days (here $b = 20$) until the original length $T$
  - recompute the statistic $B$ times; the 2.5% and 97.5% percentiles give the 95% interval
  - blocks keep the autocorrelation inside each block; i.i.d. (independent and identically distributed) resampling would destroy it and give too narrow intervals

## Jumps and Bipower Variation

- With jumps $J$ inside the day, RV collects their squares

$$\mathrm{RV}_t \xrightarrow{p} \mathrm{IV}_t + \sum_{s \in \mathbf{day}\ t} J_s^{\mathbf{2}}$$

- **Bipower variation** Barndorff-Nielsen & Shephard (2004): products of adjacent absolute returns

$$\mathrm{BV}_t = \mu_{\mathbf{1}}^{-\mathbf{2}} \frac{M}{M-1} \sum_{i=\mathbf{2}}^{M} |r_{t,i}|\,|r_{t,i-\mathbf{1}}| \xrightarrow{p} \mathrm{IV}_t, \qquad \mu_{\mathbf{1}} = \mathbb{E}|Z| = \sqrt{2/\pi}$$

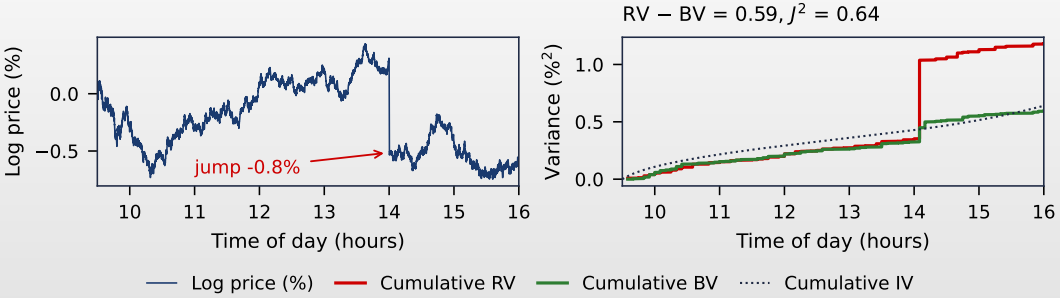
  - $Z \sim N(0,1)$, so $\mu_{\mathbf{1}}^{-\mathbf{2}} = \pi/2$; a jump enters BV (Bipower Variation) only multiplied by a small neighbouring return
  - jump part $J_t = \mathrm{RV}_t - \mathrm{BV}_t$; jump share in total variation $\sum_t J_t / \sum_t \mathrm{RV}_t$

- **Tripower quarticity**: a jump-robust estimate of $\mathrm{IQ}_t$, used by the jump test

$$\mathrm{TQ}_t = M \mu_{\mathbf{4/3}}^{-\mathbf{3}} \frac{M}{M-2} \sum_{i=\mathbf{3}}^{M} |r_{t,i}|^{\mathbf{4/3}} |r_{t,i-\mathbf{1}}|^{\mathbf{4/3}} |r_{t,i-\mathbf{2}}|^{\mathbf{4/3}}, \qquad \mu_{\mathbf{4/3}} = \mathbb{E}|Z|^{\mathbf{4/3}}$$

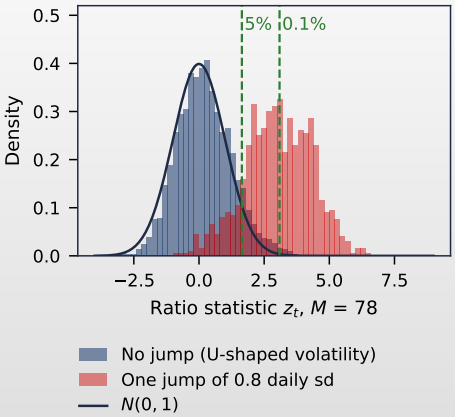
- Tiny example, $M = 4$, returns $0.1, 0.1, 0.1, 1.0$ (%)
  - $\mathrm{RV} = 1.03$; $\mathrm{BV} = \frac{\pi}{\mathbf{2}} \cdot \frac{\mathbf{4}}{\mathbf{3}} (0.01 + 0.01 + 0.10) = 0.25$; jump part $\mathrm{RV} - \mathrm{BV} = 0.78$

# A Jump Seen by RV and by BV




- Simulated day, IV $= 0.64\,\%^2$, one jump of $-0.8\%$ at 14:00, 78 five-minute returns
  - cumulative RV jumps by about $J^2 = 0.64$ at 14:00; cumulative BV keeps following IV
  - at the close $\mathrm{RV} - \mathrm{BV} = 0.59$: close to the squared jump

# The Ratio Jump Test




- $H_0$: no jump on day $t$; Huang–Tauchen ratio statistic Huang & Tauchen (2005)

$$z_t = \frac{(\mathrm{RV}_t - \mathrm{BV}_t)/\mathrm{RV}_t}{\sqrt{\theta\, M^{-1} \max\left(1,\ \mathrm{TQ}_t/\mathrm{BV}_t^2\right)}}$$

  - $(\mathrm{RV}_t - \mathrm{BV}_t)/\mathrm{RV}_t$: relative jump measure, near 0 under $H_0$
  - $\theta = \pi^2/4 + \pi - 5 = 0.609$; $\mathrm{TQ}_t/\mathrm{BV}_t^2 \geq 1$ measures how uneven volatility was in the day, floored at 1

- Under $H_0$, $z_t \approx N(0,1)$; one-sided test
  - reject when $z_t > 1.645$, 2.326, 3.09 at 5%, 1%, 0.1%; $p_t = 1 - \Phi(z_t)$, $\Phi$: standard Normal cdf
  - $N$ days tested: about $\alpha N$ rejections by chance alone (multiple testing, Chapter 9)
  - chart: with U-shaped volatility and no jump, 9% of days exceed 1.645 instead of 5%

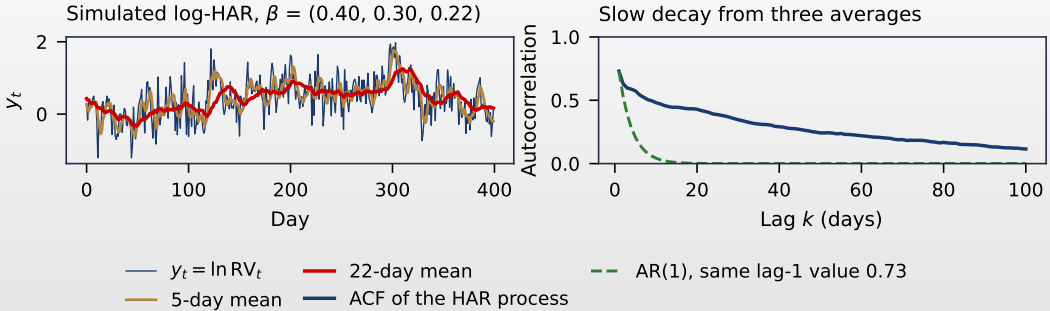
# The HAR Model

- Volatility is highly persistent, consistent with long memory (Chapter 1): the ACF (Autocorrelation Function) of $|r_t|$ decays slowly, a GARCH (Generalised Autoregressive Conditional Heteroskedasticity)(1,1) forecast reverts exponentially (Chapter 5)
- HAR (Heterogeneous Autoregressive) Corsi (2009): tomorrow's RV on three averages of past RV

$$\mathrm{RV}_{t+\mathbf{1}} = \beta_{\mathbf{0}} + \beta_d \mathrm{RV}_t + \beta_w \mathrm{RV}_t^{(\mathbf{5})} + \beta_m \mathrm{RV}_t^{(\mathbf{22})} + \varepsilon_{t+\mathbf{1}}, \qquad \mathrm{RV}_t^{(h)} = \frac{1}{h} \sum_{j=\mathbf{0}}^{h-\mathbf{1}} \mathrm{RV}_{t-j}$$

  - $\beta_d$, $\beta_w$, $\beta_m$: weights on yesterday, the last week (5 days) and the last month (22 days); $\beta_{\mathbf{0}}$: constant
  - persistence $\beta_d + \beta_w + \beta_m < 1$: shocks die out; HAR is an AR(22) whose 22 coefficients are tied to three slopes
- **log-HAR**: the same regression for $y_t = \ln V_t$
  - forecast of the level $F_{t+\mathbf{1}} = \exp(\hat{y}_{t+\mathbf{1}} + s^{\mathbf{2}}/2)$, $s^{\mathbf{2}}$: residual variance (lognormal correction)
  - positive forecasts, residuals closer to the Normal distribution
- Tiny example: $\hat{\beta} = (0.1, 0.4, 0.3, 0.2)$, $\mathrm{RV}_t = 2$, $\mathrm{RV}_t^{(\mathbf{5})} = 1.5$, $\mathrm{RV}_t^{(\mathbf{22})} = 1$
  - forecast $0.1 + 0.8 + 0.45 + 0.2 = 1.55\,\%^{\mathbf{2}}$; persistence 0.9

# Three Averages Give a Long Memory




- Simulated log-HAR, $\beta_d = 0.40$, $\beta_w = 0.30$, $\beta_m = 0.22$ (persistence 0.92)
  - left: the 22-day mean moves slowly and anchors the forecast; the daily term reacts fast
  - right: the ACF is still 0.24 at lag 50; an AR(1) with the same lag-1 value has forgotten everything by lag 15

## Inference in Time-Series Regressions

- OLS (Ordinary Least Squares) $\hat{\beta} = (X^\top X)^{-1} X^\top y$; errors autocorrelated and heteroskedastic: Newey–West (HAC) covariance

$$\hat{V}_{\mathrm{NW}} = (X^\top X)^{-1} \Big[ \hat{\Gamma}_0 + \sum_{k=1}^{q} \Big(1 - \frac{k}{q+1}\Big) (\hat{\Gamma}_k + \hat{\Gamma}_k^\top) \Big] (X^\top X)^{-1}$$

  - $\hat{\Gamma}_k = \sum_t \hat{u}_t \hat{u}_{t-k}\, x_t x_{t-k}^\top$, $\hat{u}_t$: residuals, $x_t$: regressor row; lags $q = \lfloor 4(T/100)^{2/9} \rfloor$

- **Wald test** of $q_R$ linear restrictions $H_0 : R\beta = r$

$$W = (R\hat{\beta} - r)^\top (R \hat{V} R^\top)^{-1} (R\hat{\beta} - r) \xrightarrow{d} \chi^2_{q_R}$$

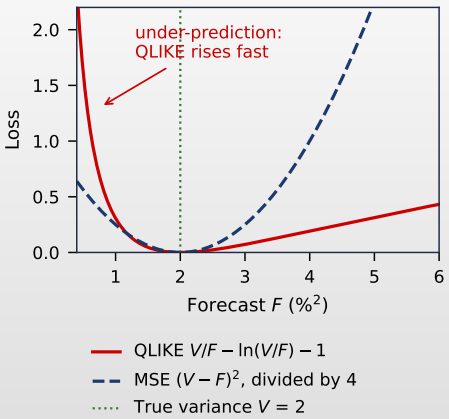
  - $R$: one row per restriction; e.g. $\beta_w = \beta_m$ is the row $(0, 0, 1, -1)$, $r = 0$

- **Ljung–Box**: are the first $h$ autocorrelations jointly zero?

$$Q(h) = T(T+2) \sum_{k=1}^{h} \frac{\hat{\rho}_k^2}{T-k} \xrightarrow{d} \chi^2_h$$

  - $\hat{\rho}_k$: sample autocorrelation at lag $k$; on residuals, a large $Q$ = memory left unexplained
  - fit: $R^2 = 1 - \sum_t \hat{u}_t^2 / \sum_t (y_t - \bar{y})^2$, the share of the variance of $y$ explained

# Loss Functions for Variance Forecasts

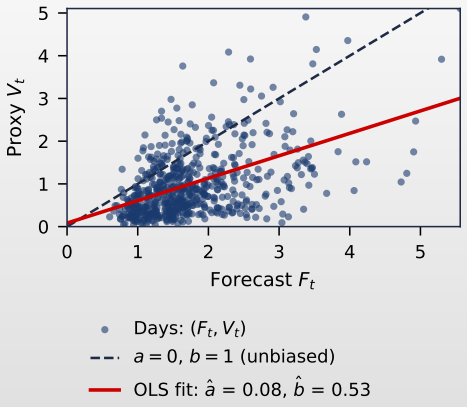



- The true variance is never observed: score $F_t$ against a proxy $V_t$, conditionally unbiased, $\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[\mathrm{IV}_t \mid \mathcal{F}_{t-1}]$
- Two losses that rank forecasts as the true variance would Patton (2011)

$$\mathrm{QLIKE}(V, F) = \frac{V}{F} - \ln \frac{V}{F} - 1, \qquad \mathrm{MSE}(V, F) = (V - F)^2$$

  - both are 0 at $F = V$; QLIKE (Quasi-Likelihood) penalises under-prediction more
- **Diebold–Mariano** Diebold & Mariano (1995): $d_t = L(V_t, F_t^A) - L(V_t, F_t^B)$
  - $t = \bar{d}/\mathrm{se}_{\mathbf{NW}}(\bar{d}) \approx N(0,1)$ under equal expected loss; $t < -1.96$: A significantly better

# Calibration: the Mincer–Zarnowitz Regression




- Regress the proxy on the forecast

$$V_t = a + b\,F_t + u_t$$

  - $a$: intercept, $b$: slope, $u_t$: error; an unbiased forecast has $a = 0$ and $b = 1$
  - $b < 1$: forecasts too extreme (chart: $\hat{b} = 0.53$, high forecasts are too high)

- Joint Wald test of $(a, b) = (0, 1)$, with $\delta = (\hat{a},\ \hat{b} - 1)^\top$

$$W = \delta^\top \hat{V}^{-1} \delta \xrightarrow{d} \chi^2_2$$

  - $\hat{V}$: $2 \times 2$ covariance of $(\hat{a}, \hat{b})$, built from the standard errors and their correlation
  - 95% confidence ellipse: the $(a, b)$ with $W \le \chi^2_{2;0.95} = 5.99$

## Benchmark Forecasts, Out of Sample

- GARCH(1,1)-$t$ on daily returns (Chapter 5): a model-based forecast

$$F_t = \sigma_t^2 = \omega + a\,\varepsilon_{t-1}^2 + b\,\sigma_{t-1}^2, \qquad \varepsilon_t = r_t - \mu = \sigma_t z_t,\ z_t \sim t_\nu$$

  - $\omega > 0$: the GARCH constant (not $\omega$, the standard deviation of microstructure noise); $a, b \geq 0$, $a + b < 1$
  - $z_t$: unit-variance Student-$t$ shocks with $\nu$ degrees of freedom

- EWMA (Exponentially Weighted Moving Average, RiskMetrics): exponentially weighted squared returns, $\lambda = 0.94$

$$\sigma_t^2 = \lambda\,\sigma_{t-1}^2 + (1-\lambda)\,r_{t-1}^2$$

  - a GARCH(1,1) with $\omega = 0$ and $a + b = 1$: no mean reversion
  - RW: yesterday's measure as today's forecast, $F_t = V_{t-1}$ (column `RW`)

- Out of sample, expanding window
  - at each origin $t-1$ every model is re-estimated on data up to $t-1$, then $F_t$ is compared with $V_t$
  - losses are averaged over the common dates of all models

## Data Used

| Series | Content | Sample |
|---|---|---|
| SPY, 5-minute bars | prices of the regular session, UTC time stamps, each bar labelled by its start | 2020-10-12 – 2026-09-18, 1 489 days |
| Bitcoin (USD), 5-minute bars | 24-hour UTC days | 659 of 747 days kept |
| SPY and Bitcoin, daily prices | adjusted closes (SPY) and closes (Bitcoin), for GARCH | since 2000 (SPY), 2014 (Bitcoin) |
| SPY daily variances (derived) | intraday RV, overnight return and total variance $V_t$ ($\%^2$, %) | one row per SPY day |
| One-day forecasts (derived) | log-HAR, HAR, GARCH(1,1)-$t$, EWMA, yesterday's RV (column `RW`) and the target $V_t$ | from 3 Jan 2022 (SPY), 1 Mar 2025 (Bitcoin) |

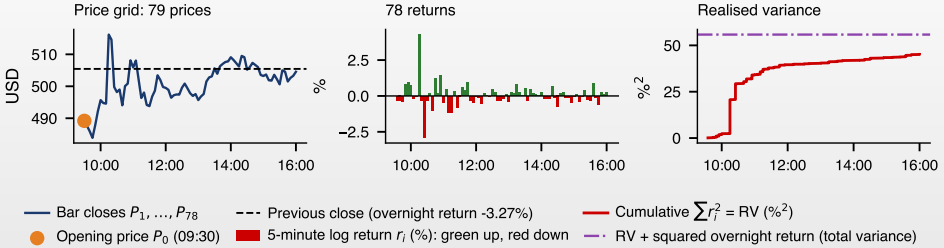
- SPY: UTC converted to New York time; bars 09:30–15:55; missing bars inside a day carry the last price; 78 returns a day, 43 on the 12 half-days
- Bitcoin: days with fewer than 95% of the 288 bars are dropped; short gaps carry the last price
- Units: returns $r = 100\,\Delta \ln P$ in %, variances in $\%^2$; annualise with 252 (SPY) or 365 (Bitcoin) days; all data end on 18 September 2026

## Setup: Run and Verify

```
# notebook section "Data" defines
# spy_intraday, btc_intraday,
# intraday_returns, spy_overnight
P = spy_intraday()         # day x 79 prices
R = intraday_returns(P)    # day x 78 returns, %
print(P.shape)
print(R.notna().sum(axis=1).value_counts())
Pb = btc_intraday()        # day x 289 prices, UTC
print(Pb.shape, Pb.index[0], Pb.index[-1])
```

- Expected output
  - (1489, 79): SPY days, from 2020-10-12 to 2026-09-18
  - 1 477 days with 78 returns, 12 with 43
  - (659, 289): Bitcoin, 2024-09-01 to 2026-09-17
- Column 0 of `P` is the opening price of the 09:30 bar; columns 1–78 are the closes of the bars
- If your counts differ, stop and check the time zone, the session filter and the cut-off date

# From Bars to Returns: One SPY Day




- 7 April 2025: opening price $P_0 = 489.19$, then bar closes $P_1 = 487.60$, $P_2 = 485.88$, …, $P_{78} = 504.54$: 79 prices give 78 returns; differencing the 78 closes alone would give 77
- $r_1 = 100 \ln(P_1/P_0) = -0.33\%$, $r_2 = -0.35\%$; $\mathrm{RV} = \sum_i r_i^2 = 45.1\,\%^2$; the overnight return $-3.27\%$ is outside the bars, total variance $V = \mathrm{RV} + r_{\text{on}}^2 = 55.8\,\%^2$
- Overnight return: the adjusted close-to-close return minus the open-to-close return, so dividends fall in the overnight part

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## Notation Used in Every Exercise

- Day $t$ has $M$ intraday returns $r_{t,i}$, $i = 1, \ldots, M$, in %
- Total daily variance $V_t = \mathrm{RV}_t + r_{\mathrm{on},t}^2$ for SPY ($r_{\mathrm{on},t}$: the overnight return); $V_t = \mathrm{RV}_t$ for Bitcoin (24 hours)
- Forecast $F_t$ of $V_t$, made at the close of day $t-1$; every loss compares $F_t$ with $V_t$

## Exercise Map

| Exercise | Question | Type | Worked example |
|---|---|---|---|
| A1 | how precise is one day's RV? | Solved | – |
| A2 | how often should a noisy price be sampled? | Solved | – |
| B1 | does SPY show a noise bias at 5 minutes? | Solved | A2 |
| B2 | which SPY days contain a jump? | Solved | – |
| A3 | can six returns reveal a jump? | Proposed | B2, A1 |
| A4 | what does HAR impose on an AR(22)? | Solved | – |
| B3 | how persistent is SPY's log variance? | Solved | A4 |
| A5 | which loss ranks variance forecasts correctly? | Solved | – |
| A6 | is GARCH less accurate, biased, or both? | Proposed | A5; Wald example |
| C1 | what is wrong in an AI answer? | Proposed | A2, B3 |
| Extension packs | A2–A5, B1–B3 extended; A7, A8, B4–B9, C2, C3 (optional, after the seminar) | Solved and Proposed | named on each task |

- **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- Part A: derivations on paper; Part B: computation and inference on SPY and Bitcoin, each ending with an interpretation question; Part C: open questions and AI critique

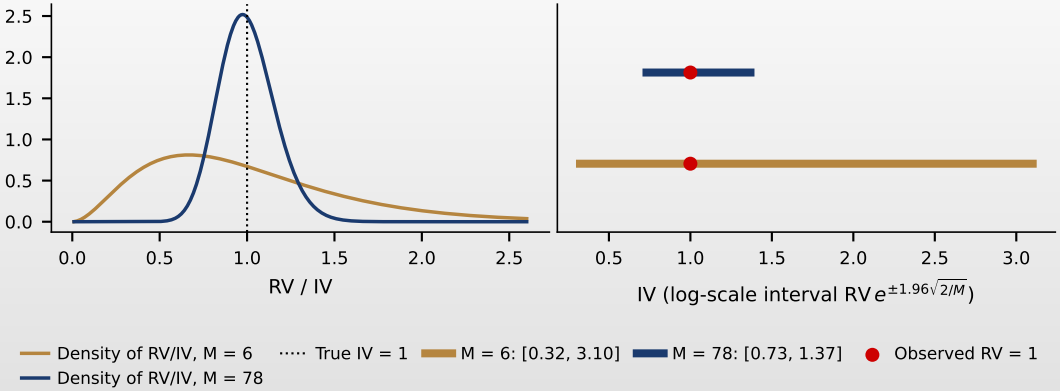
# A1 [Solved]: Precision of Realised Variance — Task

- **Question**: how precisely does one day of 5-minute returns measure integrated variance?
- **Inputs**: constant volatility, $M$ intervals of length $\Delta = 1/M$, $r_i = \sigma\sqrt{\Delta}\, Z_i$ with $Z_i$ i.i.d. $N(0,1)$, $\mathrm{IV} = \sigma^2$; use $\mathbb{E}Z^2 = 1$, $\mathrm{Var}(Z^2) = 2$, $\mathbb{E}Z^4 = 3$; $\chi^2_M$: chi-square with $M$ degrees of freedom (a sum of $M$ squared i.i.d. $N(0,1)$); no data file
- **Tasks**
  1. Show that $\mathrm{RV} = \sigma^2 \chi^2_M / M$ and derive $\mathbb{E}[\mathrm{RV}]$ and $\mathrm{Var}(\mathrm{RV})$.
  2. Show that $\mathrm{RQ} = \frac{M}{3}\sum_i r_i^4$ is unbiased for $\mathrm{IQ} = \sigma^4$ and write the estimated standard error of RV.
  3. Use the delta method for $\ln \mathrm{RV}$ and write the 95% interval for IV built on the log scale.
  4. Compute the interval factors $e^{\pm 1.96\sqrt{2/M}}$ for $M = 78$ and $M = 6$, and the relative width for $M = 78$.
- **Report**: the formulas, the two pairs of factors and one sentence on how precise one day's RV is
- Notebook: section A1

## A1 [Solved]: Solution (1/3) — Derivation

- $\mathrm{RV} = \sum_i \sigma^2 \Delta Z_i^2 = \sigma^2 \frac{1}{M} \sum_i Z_i^2 = \sigma^2 \chi^2_M / M$, so $\mathbb{E}[\mathrm{RV}] = \sigma^2$ (unbiased) and $\mathrm{Var}(\mathrm{RV}) = \sigma^4 \cdot 2M/M^2 = 2\sigma^4/M$
- $\mathbb{E}[r_i^4] = \sigma^4 \Delta^2 \mathbb{E} Z^4 = 3\sigma^4/M^2$, so $\mathbb{E}[\mathrm{RQ}] = \frac{M}{3} \cdot M \cdot 3\sigma^4/M^2 = \sigma^4$; hence $\widehat{\mathrm{Var}}(\mathrm{RV}) = 2\,\mathrm{RQ}/M$ and $\mathrm{se}(\mathrm{RV}) = \sqrt{\frac{2}{3} \sum_i r_i^4}$
- Delta method with $g(x) = \ln x$, $g'(x) = 1/x$: $\mathrm{se}(\ln \mathrm{RV}) \approx \mathrm{se}(\mathrm{RV})/\mathrm{RV} = \sqrt{2\,\mathrm{RQ}/M}/\mathrm{RV}$
- 95% interval: $\mathrm{RV} \exp(\pm 1.96\, \mathrm{se}(\ln \mathrm{RV}))$, always positive and asymmetric in levels; with constant volatility $\mathrm{RQ}/\mathrm{RV}^2 \approx 1$, so the factors are $e^{\pm 1.96\sqrt{2/M}}$
- For volatility $\sqrt{\mathrm{RV}}$, $g'(x) = 1/(2\sqrt{x})$ halves the relative standard error: the interval for volatility is narrower than the one for variance

# A1 [Solved]: Solution (2/3) — Density of RV/IV and Interval Factors




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# A1 [Solved]: Solution (3/3) — Simulated Coverage and Interpretation

- $M = 78$: $\sqrt{2/78} = 0.160$, factors $[0.73, 1.37]$: IV lies between $0.73\,\mathrm{RV}$ and $1.37\,\mathrm{RV}$, a total relative width of 0.64
- $M = 6$ (six equal intervals over the observation period): factors $[0.32, 3.10]$: almost uninformative
- Simulated coverage of the nominal 95% interval with RQ (200 000 days): 94.2% for $M = 78$, 86.7% for $M = 6$: the 95% is an asymptotic statement
- With stochastic volatility the same studentised limit holds asymptotically, for a continuous price and negligible noise, because the convergence is stable (Chapter 9)
- **Interpretation**: one day of 5-minute data pins IV down to roughly $-27\%/+37\%$

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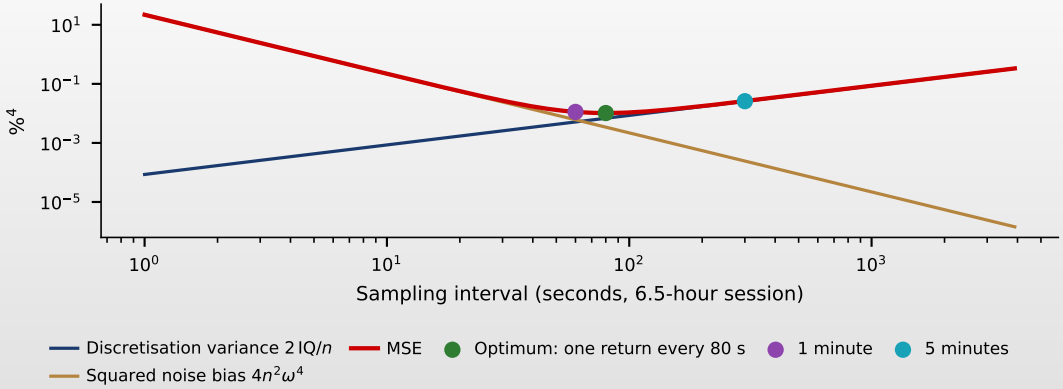
# A2 [Solved]: Noise Bias and the Optimal Frequency — Task

- **Question**: how often should we sample a price that contains noise?
- **Inputs**: the noise model of Noise and frequency; zero drift, constant volatility; $\mathrm{IV} = 1\,\%^2$, IQ $= 1$, $\omega = 0.01\%$, so $\omega^2 = 0.0001\,\%^2$; session of 23 400 seconds
- **Tasks**
  1. Write the observed return, expand $\mathbb{E}[r_i^2]$ and derive $\mathbb{E}[\mathrm{RV}^{(n)}] = \mathrm{IV} + 2n\omega^2$ for $n$ returns.
  2. Derive $\mathrm{cov}(r_i, r_{i-1}) = -\omega^2$.
  3. From MSE($n$) (slide Noise and frequency; Bandi & Russell (2008)), derive the MSE-optimal $n^*$ and its sampling interval.
  4. Fill a table for 5 minutes, 1 minute, 1 second and $n^*$: $n$, bias $2n\omega^2$, $\mathbb{E}[\mathrm{RV}]$, MSE.
- **Report**: the derivations, the table and which interval you would use in practice, with the reason
- Notebook: section A2

## A2 [Solved]: Solution (1/3) — Derivation

- $r_i = p_i^{\text{obs}} - p_{i-1}^{\text{obs}} = r_i^* + \varepsilon_i - \varepsilon_{i-1}$
- $\mathbb{E}[r_i^2] = \mathbb{E}[r_i^{*2}] + \mathbb{E}[\varepsilon_i^2] + \mathbb{E}[\varepsilon_{i-1}^2] = \mathbb{E}[r_i^{*2}] + 2\omega^2$ (cross terms have mean 0 by independence); summing over $n$ returns: $\mathbb{E}[\mathrm{RV}^{(n)}] = \mathrm{IV} + 2n\omega^2$
- Adjacent returns share only $\varepsilon_{i-1}$, with opposite signs: $\mathrm{cov}(r_i, r_{i-1}) = \mathrm{cov}(-\varepsilon_{i-1}, \varepsilon_{i-1}) = -\omega^2$
- $\frac{d}{dn}\mathrm{MSE} = -2\,\mathrm{IQ}/n^2 + 8n\omega^4 = 0$ gives $n^3 = \mathrm{IQ}/(4\omega^4)$, $n^* = (\mathrm{IQ}/(4\omega^4))^{1/3}$; the second derivative is positive: a minimum
- Numbers: $n^* = (1/(4 \times 10^{-8}))^{1/3} = 292$, one return every $23\,400/292 = 80$ seconds

# A2 [Solved]: Solution (2/3) — MSE against the Sampling Interval




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# A2 [Solved]: Solution (3/3) — Bias and MSE at Four Frequencies

| Interval | $n$ | $2n\omega^2$ | $\mathbb{E}[\mathrm{RV}]$ | MSE |
|---|---|---|---|---|
| 5 minutes | 78 | 0.0156 | 1.0156 | 0.0259 |
| 1 minute | 390 | 0.0780 | 1.0780 | 0.0112 |
| 1 second | 23 400 | 4.6800 | 5.6800 | 21.9025 |
| $n^*$, every 80 s | 292 | 0.0585 | 1.0585 | 0.0103 |

- $\mathrm{MSE} = 2/n + 4n^2 \times 10^{-8}$ in $\%^4$; at one second RV is almost six times IV
- In this model 1 minute is close to the optimum; 5 minutes has 2.5 times the minimal MSE
- **Interpretation**: the usual 5-minute choice rests on arguments outside this model
  - $\omega$ is uncertain; real noise is not i.i.d.
  - 5-minute RV is hard to beat empirically Liu, Patton & Sheppard (2015)

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# B1 [Solved]: Which Sampling Frequency? — Task

- **Prerequisites and examples**: A2 [Solved]
- **Questions**
  - does SPY show a measurable noise bias at 5 minutes?
  - which grid gives the most useful daily measure?
- **Data**: the SPY price matrix `P` of the setup (1 489 days); grid of $k \times 5$ minutes, $k = 1, 3, 6, 13$ (5, 15, 30, 65 minutes): prices at columns $o, o+k, o+2k, \ldots$ plus the open and the close, for offsets $o = 0, \ldots, k-1$ (helper `sparse_points`)
- **Tasks**
  1. Compute daily RV on one grid ($o = 0$) and the average over the $k$ shifted grids, and report $\sqrt{252\,\overline{\mathrm{RV}}}$ for each interval.
  2. For each interval compute the ratio of mean one-grid RV to mean 5-minute RV and a 95% moving-block bootstrap interval (Grids and bootstrap): 20-day blocks of the paired daily series, 2 000 resamples.
  3. Compute the correlation of today's log RV (one grid and averaged grids) with tomorrow's 5-minute log RV.
- **Report**: one table (volatility, ratio [CI (Confidence Interval)], correlation per interval) and one chart
- **Interpretation**
  - is there a measurable noise bias at 5 minutes?
  - why a block bootstrap and not an i.i.d. one?
  - which grid is most useful for forecasting?
- Notebook: section B1

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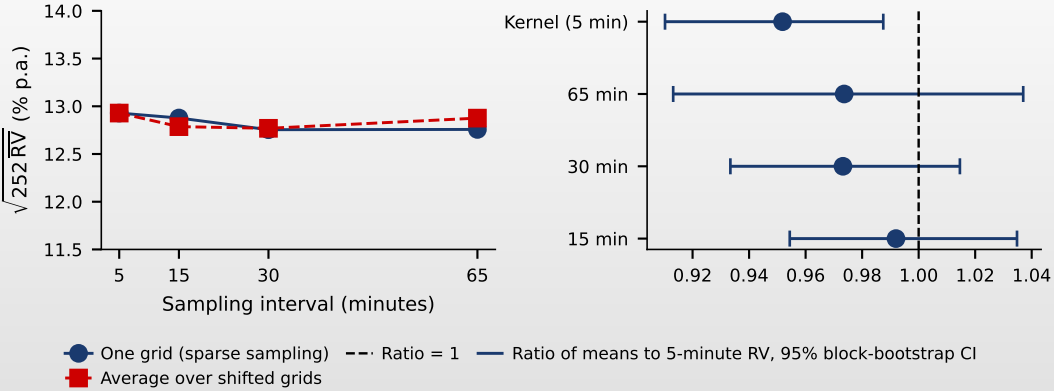
# B1 [Solved]: Solution (1/3) — Procedure and Table

- Procedure: for each day take the grid prices, compute $\mathrm{RV}^{(o)} = \sum (100\,\Delta \ln P)^2$ on each offset $o$, average over $o$; the bootstrap resamples the pairs $(\mathrm{RV}_t^{(k)}, \mathrm{RV}_t^{(5)})$ in 20-day blocks and recomputes $\sum \mathrm{RV}^{(k)} / \sum \mathrm{RV}^{(5)}$

| Interval | Vol. (one grid) | Vol. (avg. grids) | Ratio [95% CI] | Corr. next day | Corr. (avg.) |
|---|---|---|---|---|---|
| 5 min | 12.93 | 12.93 | 1.000 [1.000, 1.000] | 0.73 | 0.73 |
| 15 min | 12.88 | 12.79 | 0.992 [0.954, 1.035] | 0.71 | 0.72 |
| 30 min | 12.75 | 12.77 | 0.973 [0.933, 1.015] | 0.70 | 0.72 |
| 65 min | 12.76 | 12.88 | 0.974 [0.913, 1.037] | 0.64 | 0.70 |

- Block bootstrap: daily RVs are strongly autocorrelated; i.i.d. resampling would destroy this dependence and make the interval too narrow

# B1 [Solved]: Solution (2/3) — Signature Plot and Kernel Ratios




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# B1 [Solved]: Solution (3/3) — Noise Bias and Predictive Correlation

- All intervals contain 1: the coarse grids detect no bias relative to 5-minute RV; the signature is flat
- The predictive correlation falls from 0.73 (5 minutes) to 0.64 (65 minutes, one grid): sparse RV is noisier; averaging the shifted grids recovers part of it (0.70)
- **Interpretation**: 5-minute RV is the most useful measure here; in the kernel comparison (extension pack) the realised kernel averages 0.952 of the 5-minute RV (95% CI $[0.910, 0.987]$): a small noise bias at 5 minutes cannot be ruled out

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# B2 [Solved]: Jump Days in SPY — Task

- **Questions**
  - on which SPY days does the ratio test reject?
  - how many rejections would chance alone produce?
- **Data**: the SPY return matrix `R` (1 489 days; $M_t = 78$, or 43 on half-days); ratio test Huang & Tauchen (2005) (Bipower variation and Ratio jump test); helper `jump_test`
- **Tasks**
  1. For every day compute RV, BV, TQ, $z_t$ and the one-sided $p_t = 1 - \Phi(z_t)$; work 27 August 2021 step by step.
  2. Count the days that reject at 5%, 1% and 0.1% and compare each count with $\alpha N$.
  3. On the 0.1% days find the interval with the largest $|r_{t,i}|$ and plot the histogram of its end time.
  4. Compute the jump share $\sum_t J_t / \sum_t \mathrm{RV}_t$, with $J_t = \mathrm{RV}_t - \mathrm{BV}_t$ on 0.1% days and 0 otherwise.
- **Report**: the worked day, a table of counts against $\alpha N$, the histogram and the jump share
- **Interpretation**
  - are there more rejections than the level alone would produce?
  - what could explain them?
- Notebook: section B2

## B2 [Solved]: Solution (1/3) — One Day Step by Step

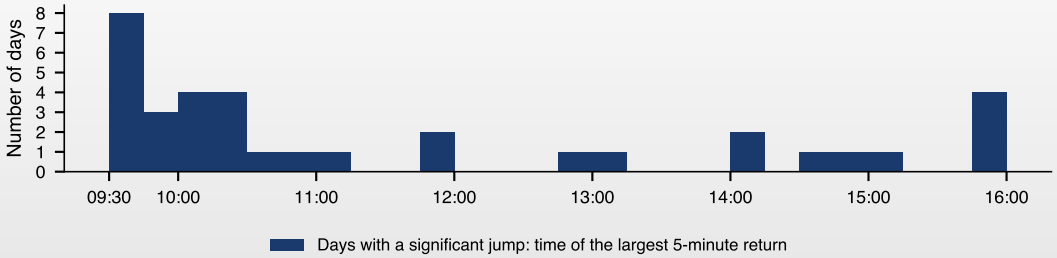
- 27 August 2021, $M = 78$: $\mathrm{RV} = 0.1354$, $\mathrm{BV} = 0.0687$, $\mathrm{TQ} = 0.00539$ (%$^2$, %$^2$, %$^4$)
- $(\mathrm{RV} - \mathrm{BV})/\mathrm{RV} = 0.493$; $\mathrm{TQ}/\mathrm{BV}^2 = 1.143 > 1$, so the floor does not bind
- $z = 0.493/\sqrt{0.609 \times 1.143/78} = 5.22$, $p = 9.0 \times 10^{-8}$: reject at 0.1%
- The largest return, $+0.24$% in the interval ending 10:05, is 44% of RV: 10:00–10:05 is when the Federal Reserve chair's Jackson Hole speech was released
- The same four steps, vectorised over days, are `jump_test(R)`

# B2 [Solved]: Solution (2/3) — Counts and Jump Share

| Level | Rejections | $\alpha N$ |
|---|---|---|
| 5% | 235 | 74.5 |
| 1% | 110 | 14.9 |
| 0.1% | 35 | 1.5 |

- $z_t$ has mean 0.50 and standard deviation 1.14 instead of 0 and 1: far more rejections than the level implies
- Jump share of total RV with 0.1% days: 2.8%
- Explanations: real jumps, and the intraday U-shape, which biases BV down (extension pack: FDR (False Discovery Rate) control and the periodicity correction)

# B2 [Solved]: Solution (3/3) — When Do the Largest Returns Occur?




- Time of the largest $|r|$ on flagged days: it peaks after the open; attributing a flag to news needs separate evidence

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## A Large Return and Evidence of a Jump

- **What do you think**: does one large return on a calm day prove a jump?
- **Answer**: no; the jump test asks whether a continuous price is implausible, given the sampling frequency
- The test statistic cannot exceed $\sqrt{M/\theta}$: $\mathrm{BV} \geq 0$ and the maximum in the denominator is at least 1
- Compare this cap with the critical value before interpreting a non-rejection
- RV, BV and tripower quarticity use the same return unit; the test statistic does not depend on it

# A3 [Proposed]: Six Returns and One Jump — Task

- **Prerequisites and examples**: B2 [Solved] (one day step by step) and A1
- **Question**: can six 5-minute returns reveal a jump?
- **Inputs**: the returns in the table (%); the last one is a news jump; $\mathrm{TQ} = 0.0097$ (supplied); $\theta = 0.609$; critical value at 0.1%: 3.09; formulas: Bipower variation and Ratio jump test
- **Tasks**
  1. Complete the table and compute RV and BV with $M = 6$.
  2. Compute the jump part $\mathrm{RV} - \mathrm{BV}$ and its share of RV.
  3. Compute $\mathrm{TQ}/\mathrm{BV}^2$, apply the floor $\max(1, \mathrm{TQ}/\mathrm{BV}^2)$ and compute the ratio statistic $z$.
  4. Decide at 0.1% and compare $z$ with its largest possible value $\sqrt{M/\theta}$.
- **Report**: the completed table, RV, BV, $z$, the decision and one sentence of interpretation

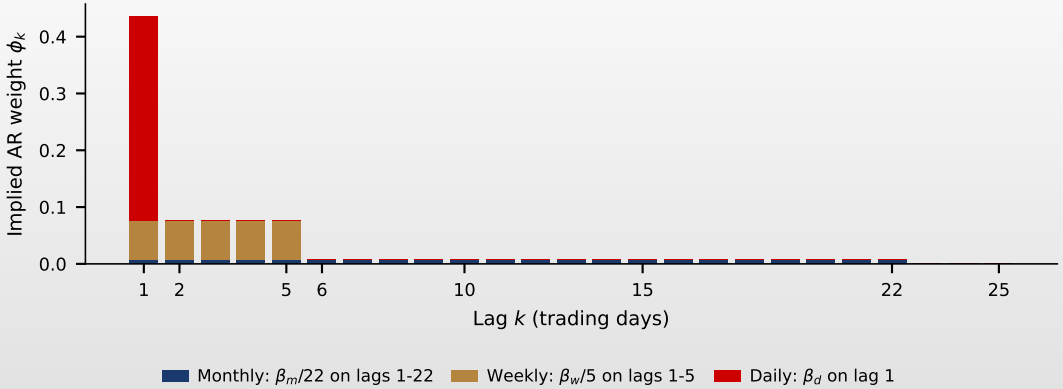
| $i$ | $r_i$ | $r_i^2$ | $\lvert r_i r_{i-1} \rvert$ |
|---|---|---|---|
| 1 | 0.08 | 0.0064 | – |
| 2 | −0.15 | | |
| 3 | 0.03 | | |
| 4 | 0.10 | | |
| 5 | −0.04 | | |
| 6 | −0.90 | | |

# A4 [Solved]: HAR as a Restricted AR(22) — Task

- **Question**: which restrictions does HAR impose on an AR(22) for log variance?
- **Inputs**: the log-HAR of The HAR model Corsi (2009), for $y_t = \ln V_t$: daily, weekly (5-day) and monthly (22-day) means of $y$; rounded SPY estimates $\hat{\beta} = (-0.08, 0.36, 0.34, 0.17)$
- **Tasks**
  1. Expand the two means and write the implied AR coefficients $\phi_1$, $\phi_2 = \ldots = \phi_5$ and $\phi_6 = \ldots = \phi_{22}$.
  2. With $\theta = (\phi_0, \phi_1, \ldots, \phi_{22})$, write two rows of the matrix $R$ in $R\theta = 0$ and count the restrictions.
  3. Compute $\phi_1$, $\phi_{2..5}$, $\phi_{6..22}$ and the persistence $\sum_k \phi_k$ for $\hat{\beta}$, and draw the 22 weights.
- **Report**: the three weight formulas, two rows of $R$, the numbers and what the persistence implies for shocks
- Notebook: section A4

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# A4 [Solved]: Solution (1/2) — Implied AR(22) Weights




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# A4 [Solved]: Solution (2/2) — Restrictions and Persistence

- Lag 1 is in all three components: $\phi_{\mathbf{1}} = \beta_d + \beta_w/5 + \beta_m/22$; lags 2–5 in the weekly and monthly ones: $\phi_k = \beta_w/5 + \beta_m/22$; lags 6–22 only in the monthly one: $\phi_k = \beta_m/22$
- Rows of $R$ (column 0: intercept): $(0, 0, 1, -1, 0, \ldots, 0)$ for $\phi_{\mathbf{2}} = \phi_{\mathbf{3}}$; $(0, \ldots, 0, 1, -1, 0, \ldots)$ at columns 6, 7 for $\phi_{\mathbf{6}} = \phi_{\mathbf{7}}$
- Count: 3 equalities among $\phi_{\mathbf{2}}, \ldots, \phi_{\mathbf{5}}$ plus 16 among $\phi_{\mathbf{6}}, \ldots, \phi_{\mathbf{22}}$: 19 rows, $R$ is $19 \times 23$ (22 lags minus 3 free slopes)
- $\phi_{\mathbf{1}} = 0.36 + 0.34/5 + 0.17/22 = 0.436$; $\phi_{\mathbf{2..5}} = 0.076$; $\phi_{\mathbf{6..22}} = 0.008$; $\sum_k \phi_k = \beta_d + \beta_w + \beta_m = 0.87$
- **Interpretation**: persistence $0.87 < 1$: shocks to log variance decay slowly but die out; the data test of the 19 restrictions is in the extension pack

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# B3 [Solved]: log-HAR for SPY with Robust Inference — Task

- **Prerequisites and examples**: A4 [Solved]
- **Questions**
  - how persistent is SPY's log total variance?
  - do robust standard errors change any conclusion?
- **Data**: SPY total daily variance from the setup, $V_t = \mathrm{RV}_t + r^2_{\mathrm{on},t}$, 1 489 days; regressors dated $t-1$: $\ln V_{t-1}$ and the means of $\ln V$ over days $t-5..t-1$ and $t-22..t-1$ (mean of logs); the first 22 days are lost
- **Tasks**
  1. Build the design matrix and check your row for 8 April 2025 against the solution.
  2. Estimate log-HAR by OLS; report the usual and the Newey–West (HAC) standard errors with 7 lags, the rule of Regression inference.
  3. Test $\beta_w = \beta_m$ with a Wald test using the Newey–West covariance.
  4. Run Ljung–Box tests with 22 lags on $\ln V_t$ and on the residuals, plot both ACFs, and compare $R^2$ with an AR(1).
- **Report**: the coefficient table with both standard errors, the tests with $p$-values and the chart
- **Interpretation**
  - which component matters most?
  - do robust errors change a conclusion?
  - has HAR captured the persistence?
- Notebook: section B3

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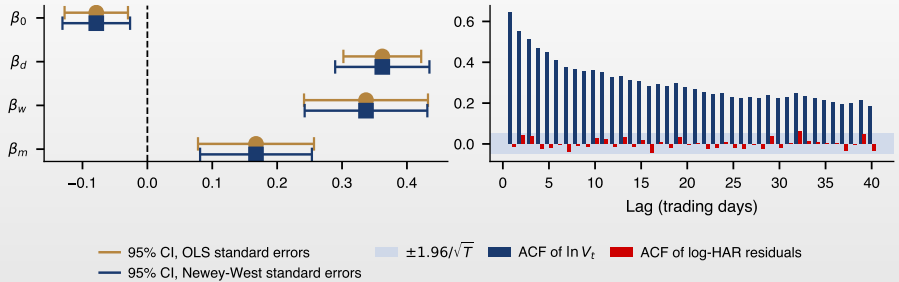
# B3 [Solved]: Solution (1/2) — Design Row and Estimates

- Row for 8 April 2025: $y_t = \ln V_t = 2.976$; regressors $\ln V_{t-\mathbf{1}} = 4.022$ (7 April), weekly mean 2.068, monthly mean 0.612
- Fitted value $\hat{y}_t = 2.177$: the model expects a partial return towards the monthly level
- Usable observations: $1\,489 - 22 = 1\,467$; $R^{\mathbf{2}} = 0.470$ against 0.413 for an AR(1) ($\hat{\phi} = 0.643$); persistence 0.866

| | $\hat{\beta}$ | SE OLS | SE NW | $t_{\mathbf{NW}}$ |
|---|---|---|---|---|
| $\beta_{\mathbf{0}}$ | −0.079 | 0.025 | 0.027 | −3.0 |
| $\beta_d$ | 0.362 | 0.031 | 0.037 | 9.8 |
| $\beta_w$ | 0.337 | 0.049 | 0.048 | 7.0 |
| $\beta_m$ | 0.167 | 0.046 | 0.044 | 3.8 |

- Newey–West errors are up to 20% larger than the OLS ones; all slopes stay significant

# B3 [Solved]: Solution (2/2) — Tests and Chart




- Wald $\beta_w = \beta_m$: 4.17, $p = 0.041$: rejected at 5%, the weekly weight is larger
- Ljung–Box(22): ln $V_t$ 4 641, residuals 17.5 ($p = 0.73$): three coefficients absorb the persistence; first-order residual autocorrelation $-0.014$
- **Interpretation**: the weekly and daily components dominate; robust errors do not change a conclusion; HAR leaves no significant autocorrelation

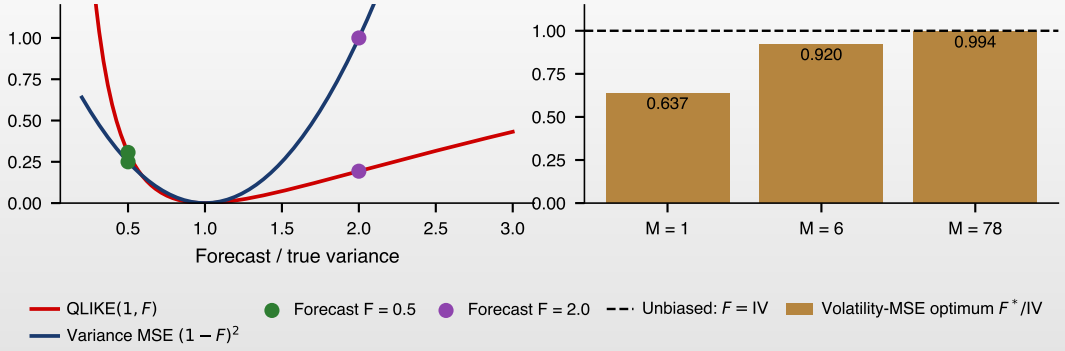
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# A5 [Solved]: Which Loss Ranks Variance Forecasts? — Task

- **Question**: which loss ranks variance forecasts correctly when the target is a noisy proxy?
- **Inputs**: the conditionally unbiased proxy $V_t$ and the losses QLIKE and variance MSE of Forecast losses Patton (2011)
- **Tasks**
  1. Minimise $\mathbb{E}[\text{QLIKE}(V_t, F) \mid \mathcal{F}_{t-1}]$ over $F$ and show that the minimiser is $\mathbb{E}[\text{IV}_t \mid \mathcal{F}_{t-1}]$.
  2. For two forecasts $F_A$, $F_B$ show that $\Delta L = V(1/F_A - 1/F_B) + \ln(F_A/F_B)$ is linear in $V$, so its conditional mean is the same with $V_t$ or with $\text{IV}_t$: the ranking is preserved.
  3. With true variance 1, compute QLIKE and variance MSE for $F = 0.5$ and $F = 2$ and compare the two rankings.
- **Report**: the first-order condition, the ranking argument, four loss values and which loss you would use
- Notebook: section A5

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# A5 [Solved]: Solution (1/2) — QLIKE and MSE Loss Curves



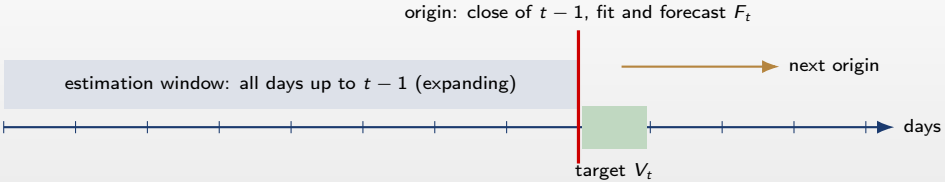

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# A5 [Solved]: Solution (2/2) — First-Order Condition and Ranking

- $\partial_F \mathbb{E}[V/F - \ln V + \ln F - 1] = -\mathbb{E}V/F^2 + 1/F = 0$ gives $F = \mathbb{E}[V_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[\mathrm{IV}_t \mid \mathcal{F}_{t-1}]$; second derivative $1/F^2 > 0$
- $\mathrm{QLIKE}(V, F_A) - \mathrm{QLIKE}(V, F_B) = V(1/F_A - 1/F_B) + \ln(F_A/F_B)$: linear in $V$, and $F_A$, $F_B$ are known at $t-1$, so replacing $V_t$ by the conditionally unbiased $\mathrm{IV}_t$ leaves $\mathbb{E}[\Delta L \mid \mathcal{F}_{t-1}]$ unchanged
- QLIKE: $F = 0.5$ gives $2 - \ln 2 - 1 = 0.307$, $F = 2$ gives $0.5 + \ln 2 - 1 = 0.193$; variance MSE: 0.25 and 1, the opposite ranking
- Chart, right: the MSE-on-volatility optimum $F^*/\mathrm{IV}$ for $M = 1, 6, 78$ (A5 Extended)
- **Interpretation**: both losses are robust, but QLIKE penalises under-prediction more, which matters for risk; MSE on *volatility* is not robust (extension pack)

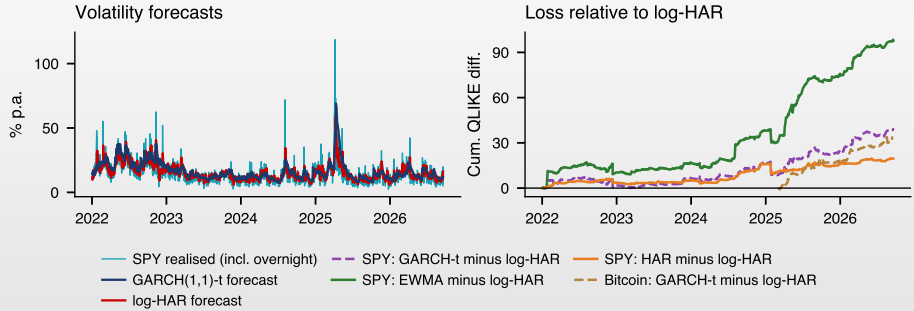
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## Forecast Evaluation: Origin, Target, Window




- The out-of-sample scheme of Out-of-sample forecasts: no value from day $t$ enters the forecast $F_t$
- SPY: 1 180 origins from 3 January 2022 to 18 September 2026

# One-Day Forecasts and Cumulative Losses




- Mean QLIKE: log-HAR 0.310, GARCH(1,1)-$t$ 0.343; a rising cumulative line means log-HAR has the lower loss
- $d_t$ = QLIKE(log-HAR) − QLIKE(GARCH) is the input of A6: is the gap more than chance?

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## A Joint Calibration Test: A Numerical Example

- Toy Mincer–Zarnowitz estimates: $\hat{a} = 0.10$, $\mathrm{SE}(\hat{a}) = 0.05$, $\hat{b} = 0.90$, $\mathrm{SE}(\hat{b}) = 0.08$, correlation $\rho = -0.5$. Test $a = 0$, $b = 1$.
- The covariance is $-0.5(0.05)(0.08) = -0.002$. Standardised deviations are $t_a = 2$, $t_b = -1.25$.
- $W = (t_a^2 + t_b^2 - 2\rho t_a t_b)/(1 - \rho^2) = (4 + 1.5625 - 2.5)/0.75 = 4.0833$. Against $\chi^2_2$, $p = e^{-W/2} = 0.130$: do not reject jointly at 5%.
- If $|\rho|$ is near one, the covariance matrix is nearly singular: rounding can change the conclusion. In MZ (Mincer–Zarnowitz), $b > 1$ means realised variance varies more than the forecasts in the calibration regression; these coefficients are not GARCH parameters.

# A6 [Proposed]: Accuracy versus Calibration — Task

- **Prerequisites and examples**: the preceding joint calibration example and B3 step 3
- **Question**: is GARCH less accurate than log-HAR, biased, or both?
- **Inputs**: SPY one-day forecasts, January 2022 – September 2026; $d_t$ = QLIKE(log-HAR) − QLIKE(GARCH) has mean $\bar{d} = -0.0327$, Newey–West SE (Standard Error) 0.0142; MZ regression for GARCH, $V_t = a + bF_t + u_t$: $\hat{a} = 0.2418$ (SE 0.2310), $\hat{b} = 0.7046$ (SE 0.2382), $\mathrm{corr}(\hat{a}, \hat{b}) = -0.9750$ (keep four decimals)
- **Tasks**
  1. Accuracy, $H_0$: equal expected loss. Compute the Diebold–Mariano $t$, its two-sided $p$-value, and decide at 5%.
  2. Calibration, $H_0$: $a = 0$, $b = 1$. Compute the separate $t$-statistics for $a = 0$ and $b = 1$.
  3. Build the $2 \times 2$ covariance matrix, compute the joint Wald statistic $W$ of Calibration regression and its $\chi^2_2$ $p$-value.
  4. Draw the 95% confidence ellipse, mark $(0, 1)$ and explain when separate and joint tests disagree.
- **Report**: three statistics, three decisions and one sentence separating accuracy from calibration

# C1 [Proposed]: Critique an AI Answer — Task

- **Prerequisites and examples**: A2 and B3 [Solved]; a code-reading exercise, no tick data are needed
- **Setting**: a student gives an AI assistant a pandas Series `px` of clean, time-sorted SPY trade prices in New York time, 09:30–16:00, several days, in USD (so returns are decimal, not %)
- **Tasks**
  1. The answer on the next slide contains three conceptual errors; everything else is correct under this setting. Find them.
  2. For each error give the evidence: a derivation, a check on the output, or a number from the lecture or this seminar.
  3. Give the corrected formula and code line. For $RV_t = 10^{-4}$, report the annualised volatility printed by the original and corrected code.
  4. Log each error you actually found in the format of `AI_ERRORS.md`: request, answer, error, how found, correction.
- **Report**: a table with one row per error (what, evidence, correction) and the annualised volatility printed by the original and the corrected code for $\mathrm{RV}_t = 10^{-4}$
- Hints: A2 on $\mathbb{E}[\mathrm{RV}^{(n)}]$ with noise; what happens between 16:00 and 09:30; plug in $\mathrm{RV}_t = 1.0 \times 10^{-4}$ ($1\%^2$)

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# C1 [Proposed]: Critique an AI Answer — the Answer

- **Request**: "Compute the daily realised variance, the annualised volatility and the HAR regressors from px."
- **Answer, text**
  - $\mathrm{RV}_t = \sum_i r_{t,i}^2$ with $r_{t,i}$ the intraday log returns of day $t$; use every tick: more returns always make $\mathrm{RV}_t$ more precise
  - $\mathrm{RV}_t$ is SPY's daily variance: compare it directly with squared close-to-close returns and with the GARCH variance
  - HAR: regress $\mathrm{RV}_{t+1}$ on $\mathrm{RV}_t$ and on its 5-day and 22-day averages Corsi (2009)

```
import numpy as np, pandas as pd
lp = np.log(px)
r = lp.groupby(lp.index.date).diff().dropna()
RV = (r**2).groupby(r.index.date).sum()
vol_ann = np.sqrt(RV) * 252
H = pd.DataFrame({'d': RV,
                  'w': RV.rolling(5).mean(),
                  'm': RV.rolling(22).mean(),
                  'y': RV.shift(-1)}).dropna()
```

# Take-Home Choice

## Take-home: choose one

- B4, jumps in Bitcoin (model: B2), or B7, forecasting the next week (model: B3, A6)
- Write it as a Quantlet folder (code, `Metainfo.txt`, the chart): the format the team project uses
- AI use is allowed; note the errors you actually catch, in the format of `AI_ERRORS.md`
- Extension packs are optional

## Assessment criteria

- correct calculations and inference
- reproducible code
- interpretation of the results and their limits
- documented AI use

# Extension Packs: What Each One Needs

| Pack | Exercises | Needs |
|---|---|---|
| 1. Noise | A2 extended [Solved], A7, B1 extended [Solved] | A2, B1 |
| 2. Jumps | A3 extended, B2 extended [Solved], B4 | A3, B2 |
| 3. Distributions | B5 | B1 (block bootstrap) |
| 4. HAR | A4 extended [Solved], B3 extended [Solved], A8, B6 | A4, B2, B3 |
| 5. Forecasting | A5 extended [Solved], B7, B7 extended, B8 | A5, A6, the evaluation workflow |
| 6. Roughness | B9 | B1, Chapter 9 (rough volatility) |
| 7. Projects | C2, C3 | B3, A5, B8, the lecture case study |

- The notebook supplies the worked examples and computational helpers; extension tasks also require your own code. Within a pack, work in order
- C2 and C3 can seed the team project

## What Does Two-Scale Estimation Correct?

- Two-Scale Realised Variance (TSRV) subtracts an estimate of the noise bias from the variance averaged over sparse grids
- TSRV cancels the leading noise term and keeps the information of many observations
- Example: sparse grids 1.20, full grid 1.50, correction weight 0.20: $1.20 - 0.20 \times 1.50 = 0.90$
- The finite-sample factor corrects the remaining attenuation; state whether you applied it

# A2 Extended [Solved]: Estimating the Noise Variance — Task

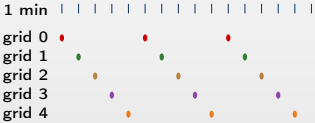
- **Questions**
  - how can $\omega^2$ be estimated from the data?
  - in which limit is each estimator consistent?
- **Inputs**: the model of A2, $n$ returns per day
- **Tasks**
  1. Show that $\mathbb{E}[\mathrm{RV}^{\mathrm{all}}/(2n)] = \omega^2 + \mathrm{IV}/(2n)$ and state its limit as $n \to \infty$.
  2. Propose a second estimator from the first-order autocovariance and state when it is consistent.
- **Report**: both derivations, the limit of each estimator and its consistency condition
- Notebook: section A2

## A2 Extended [Solved]: Solution — Estimating the Noise Variance

- $\mathbb{E}[\mathrm{RV}^{\mathsf{all}} \mid \sigma] = \mathrm{IV} + 2n\omega^2$, so $\mathbb{E}[\mathrm{RV}^{\mathsf{all}}/(2n)] = \omega^2 + \mathrm{IV}/(2n)$; a law of large numbers for the noise terms gives $\hat{\omega}^2 = \mathrm{RV}^{\mathsf{all}}/(2n) \xrightarrow{p} \omega^2$ as $n \to \infty$ Bandi & Russell (2008)
- $-\widehat{\mathrm{cov}}(r_i, r_{i-1})$ estimates $\omega^2$ (A2), consistent as the number of returns grows, provided the noise is i.i.d. and independent of the price
- In finite samples the first estimator is contaminated by $\mathrm{IV}/(2n)$; A7 compares it with the oracle value

# A7 [Proposed]: Two-Scale RV — Task

- **Prerequisites and examples**: A2 [Solved]
- **Inputs**: a synthetic illiquid stock with a stipulated 390-minute session: $\mathrm{IV} = \mathsf{IQ} = 1$, $\omega = 0.03\%$; one day: 1-minute returns ($n = 390$) give $\mathrm{RV}^{\mathbf{all}} = 1.70$; $K = 5$ shifted 5-minute grids give $\mathrm{RV}^{\mathbf{avg}} = 1.10$




1. Compute the noise bias $2n\omega^{\mathbf{2}}$ at 5 and 1 minutes, and $n^*$ of A2 with its interval.
2. With $\bar{n} = (n - K + 1)/K$, compute TSRV $= \mathrm{RV}^{\mathbf{avg}} - \frac{\bar{n}}{n}\mathrm{RV}^{\mathbf{all}}$, with and without the factor $1/(1 - \bar{n}/n)$.
3. Estimate $\omega^{\mathbf{2}}$ twice: the oracle $(\mathrm{RV}^{\mathbf{all}} - \mathrm{IV})/(2n)$ and the feasible $\mathrm{RV}^{\mathbf{all}}/(2n)$ of A2 extended.
4. Show that TSRV removes the noise bias to first order. Optional: with $\mathrm{Var}(\text{error}) \approx aK/n + bn/K^{\mathbf{2}}$, find the $K$ that balances the terms and the rate of TSRV.

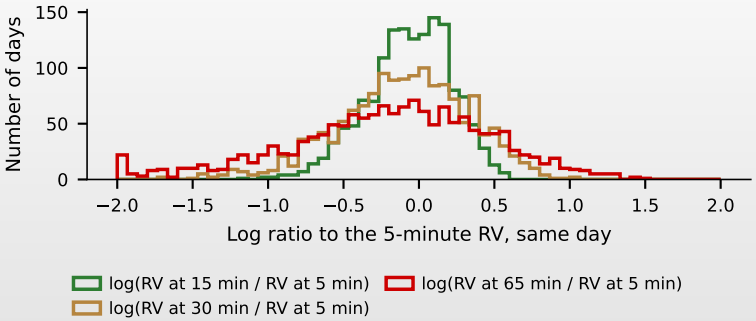
- **Report**: the biases and $n^*$, both TSRV values, both estimates of $\omega^{\mathbf{2}}$ and one sentence on when TSRV pays off

# B1 Extended [Solved]: A Realised Kernel on 5-Minute Bars — Task

- **Question**: does a noise-robust estimator change SPY's 5-minute RV?
- **Data**: SPY returns `R` and prices `P`; helper `realized_kernel` Barndorff-Nielsen et al. (2008): Parzen weights, bandwidth $H = \lceil 3.5134\, \xi^{4/5} n^{3/5} \rceil$ per day ($\lceil \cdot \rceil$: round up), noise-to-signal ratio $\xi^2 = \hat{\omega}^2 / \sqrt{\widehat{\mathrm{IQ}}}$ (in practice $\sqrt{\widehat{\mathrm{IQ}}} \approx \widehat{\mathrm{IV}}$), the course grid endpoints (open and close)
- **Tasks**
  1. Compute the kernel for every day and the ratio of its mean to mean RV, with the block bootstrap of B1.
  2. Plot the distribution of $\ln(\mathrm{RV}^{(k)}/\mathrm{RV}^{(5)})$ for 15, 30 and 65 minutes.
- **Report**: the kernel/RV ratio with its bootstrap CI, the median bandwidth $H$ and the distribution chart
- Notebook: section B1

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# B1 Extended [Solved]: Solution — A Realised Kernel on 5-Minute Bars




- Kernel / RV $= 0.952$, CI $[0.910, 0.987]$ (median bandwidth $H = 7$): the kernel adds back small negative autocorrelations ($\hat{\rho}_1 = -0.008$); significant but economically minor
- Coarser grids scatter around the 5-minute value: sparse RV adds noise; the noise-robust realised kernel is about 5% lower: a small noise bias at 5 minutes cannot be ruled out

MFM_ch9_seminar

# A3 Extended [Proposed]: Why BV Ignores a Jump — Task

- **Prerequisites and examples**: A1 [Solved]
- **Inputs**: one jump of size $\kappa$ in an **interior** interval $j$ ($1 < j < M$): $r_j = \kappa + r_j^c$; the other returns are diffusive, $O_p(\Delta^{1/2})$
- **Tasks**
  1. Show that to leading order the jump enters BV only through $|r_{j-1}|\,|\kappa| + |\kappa|\,|r_{j+1}|$, and that these products are $O_p(\Delta^{1/2})$.
  2. Conclude that $\mathrm{BV} \to \mathrm{IV}$ while $\mathrm{RV} \to \mathrm{IV} + \kappa^2$; how does a jump in the last interval (as in A3) differ?
- **Report**: the derivation, the limits of BV and RV, and the comparison with the last-interval jump in A3

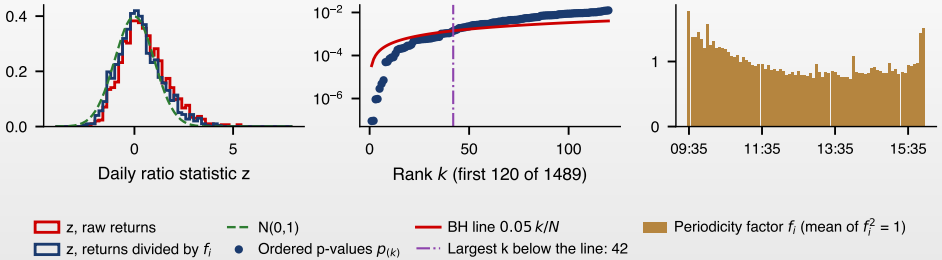
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# B2 Extended [Solved]: Multiplicity and Intraday Seasonality — Task

- **Question**: how many of B2's rejections survive a false-discovery-rate control and a correction for the intraday U-shape?
- **Data**: the $N = 1\,489$ daily $p$-values of B2; helpers `bh_count`, `periodicity_sd`
- **Tasks**
  1. Benjamini–Hochberg Benjamini & Hochberg (1995): sort the $p$-values and reject the $k$ smallest, $k$ the largest index with $p_{(k)} \le kq/N$; apply $q = 5\%$ and 1%, and Bonferroni ($p < 0.05/N$).
  2. Periodicity factor Boudt, Croux & Laurent (2011): scale returns by $\sqrt{\mathrm{BV}_t/M_t}$, take the root mean square of each 5-minute slot, normalise so that the mean of $f_i^2$ is 1; divide $r_{t,i}$ by $f_i$ and repeat B2(2) and step 1.
- **Interpretation**: BH (Benjamini–Hochberg) controls the FDR for valid $p$-values that are independent or positively dependent Benjamini & Yekutieli (2001); which count would you use to build a jump regressor?
- **Report**: the number of rejections with BH at 5% and 1% and with Bonferroni, before and after the periodicity correction, and the count you would use for a jump regressor

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# B2 Extended [Solved]: Solution — Multiplicity and Intraday Seasonality




- BH at 5%: the largest $k$ with $p_{(k)} \leq k \times 0.05/1\,489$ is $k = 42$ ($p_{(k)} = 0.00140$ against 0.00141): 42 jump days; at 1%: 9; Bonferroni: 7
- After dividing by $f_i$ ($f = 1.78$ at the open, 0.73 at the minimum): 167 rejections at 5%, 20 at 0.1%, 17 under FDR 5%
- **Interpretation**: build the jump regressor from FDR-controlled, periodicity-corrected days

MFM_ch9_seminar

# B4 [Proposed]: Jumps in Bitcoin — Task

- **Prerequisites and examples**: B2 [Solved]
- **Data**: Bitcoin price matrix `Pb`, 24-hour UTC days with at least 95% of the 288 bars, 659 days of 747; a day is labelled by its UTC date
- **Tasks**
  1. Apply the ratio test of B2 at 0.1% and build a daily table: date, $M_t$, RV, BV, $z$, weekend flag.
  2. For weekdays and weekends report the number of days, the number of rejections and their rate, and the mean $\mathrm{BV}/\mathrm{RV}$.
  3. Compute the jump share of total variation and a one-sided binomial test of the overall rate against 0.001.
- **Interpretation**: the binomial test assumes independent, correctly sized daily tests
  - what does a rejection establish?
  - what does it not establish?
  - which group rejects more often?
  - why might that group reject more often?
- **Report**: weekday/weekend table (days, rejections, rate, mean BV/RV), jump share and binomial p-value; chart of the daily jump statistic with weekends marked; four interpretation sentences

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# B5 [Proposed]: Returns Standardised by Realised Volatility — Task

- **Prerequisites and examples**: B1 [Solved] (block bootstrap)
- **Data**: SPY open-to-close returns (1 489 days) and Bitcoin 24-hour returns (659 days), each aligned with the same-day RV Andersen et al. (2001); one row: 7 April 2025, $r_t = 3.09\%$, $\mathrm{RV}_t = 45.1$, $z_t = r_t/\sqrt{\mathrm{RV}_t} = 0.46$
- **Tasks**
  1. Compute the Pearson kurtosis $\mathbb{E}(x - \bar{x})^4/s^4$ of $u_t = r_t/\mathrm{sd}(r)$ and of $z_t$, with 95% intervals from a paired moving-block bootstrap (20-day blocks, 2 000 resamples).
  2. Plot histograms and Normal quantile–quantile (QQ) plots of $u_t$ and $z_t$; run Jarque–Bera (JB) on $z_t$; compute the share of $|z_t| > 3$ (Normal: 0.27%).
- **Interpretation**: where do the heavy tails come from? Optional benchmark: for i.i.d. Gaussian intraday returns $z_t$ has kurtosis $3M/(M+2)$
- **Report**: the distribution summaries, requested test statistics and bootstrap intervals, with an interpretation of Normality

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# A4 Extended [Solved]: The Wald Test of the HAR Restrictions — Task

- **Prerequisites and examples**: A4 [Solved]
- **Tasks**
  1. Write the Wald statistic for $R\theta = 0$ with a Newey–West covariance $\hat{V}_{\text{NW}}$ and give its limiting distribution under HAR.
  2. Explain why a HAC covariance is needed for the log-variance regression.
- **Report**: the Wald statistic, its limiting distribution and the reason for the HAC covariance

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## A4 Extended [Solved]: Solution — The Wald Test of the HAR Restrictions

- $W = (R\hat{\theta})^\top (R\hat{V}_{\text{NW}} R^\top)^{-1} R\hat{\theta} \to \chi^2_{19}$ under the 19 HAR restrictions
- The errors of the log-variance regression are heteroskedastic and may be autocorrelated: the usual OLS covariance would misstate $\text{Var}(R\hat{\theta})$
- The data test is B3 extended, step 1

## What Changes in HAR Extensions?

- HARQ adds realised quarticity to HAR, for measurement uncertainty
- HAR-CJ separates continuous variation (C) from jumps (J)
- Semivariance HAR (SHAR) separates the variation from positive returns from that from negative returns
- The two semivariances add up to realised variance

# B3 Extended [Solved]: The AR(22) Test and HARQ — Task

- **Prerequisites and examples**: A4 extended, B3 [Solved]
- **Data**: as in B3; realised quarticity $\mathrm{RQ}_t$ from the intraday returns (it does not cover the overnight return)
- **Tasks**
  1. Estimate the unrestricted AR(22) of $\ln V_t$ and test the 19 HAR restrictions with a Newey–West Wald test.
  2. Estimate HARQ in levels Bollerslev, Patton & Quaedvlieg (2016),
     $V_{t+\mathbf{1}} = \beta_{\mathbf{0}} + (\beta_d + \beta_Q \sqrt{\mathrm{RQ}_t}) V_t + \beta_w V_t^{(w)} + \beta_m V_t^{(m)} + \varepsilon$, on intraday RV and on total variance.
  3. Forecast from 3 January 2022 on an expanding window, replacing forecasts that fall outside the [min, max] range of the in-sample values by the sample mean Bollerslev, Patton & Quaedvlieg (2016), and run DM (Diebold–Mariano) against HAR in levels under QLIKE.
- **Report**: $W$ and its p-value, $\hat{\beta}_Q$ with standard errors on both targets, QLIKE for HARQ and HAR, and the DM test
- Notebook: section B3

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## B3 Extended [Solved]: Solution — The AR(22) Test and HARQ

- (1) $W = 18.1$, $p = 0.51$ ($\chi^2_{19}$); $R^2$ 0.477 (AR(22)) vs 0.470 (HAR): the restrictions are not rejected; HAR saves 19 parameters
- (2) $\hat{\beta}_Q = -0.0088$ (SE 0.0042) on intraday RV, significant at 5%; $-0.0022$ (SE 0.0024) on total variance, not significant
- (3) QLIKE 0.344 (HARQ) vs 0.326 (HAR), DM $t = 1.23$, $p = 0.22$: no significant difference
- **Interpretation**: the negative sign expected for $\beta_Q$ (attenuation from measurement error, A8) appears, but with 5-minute RV the measurement error is small and the overnight return, not covered by RQ, dominates the noise in total variance

# A8 [Proposed]: Attenuation and the Case for HARQ — Task

- **Prerequisites and examples**: A4 [Solved], B3 extended
- **Inputs**: $\mathrm{RV}_t = V_t^* + u_t$ (latent signal plus error), $V_t^*$ the intraday integrated variance (IV), distinct from total variance $V_t$ including overnight; $\mathbb{E}[u_t \mid \sigma] = 0$, $\mathrm{Var}(u_t \mid \sigma) = 2\,\mathrm{IQ}_t/M$ (asymptotic, A1); errors serially uncorrelated and uncorrelated with $V^*$ and $\eta$; $V^*_{t+\mathbf{1}} = \beta_\mathbf{0} + \beta V_t^* + \eta_{t+\mathbf{1}}$, $\beta > 0$
- **Tasks**
  1. Using plim $\hat{\beta} = \mathrm{cov}(\mathrm{RV}_{t+\mathbf{1}}, \mathrm{RV}_t)/\mathrm{Var}(\mathrm{RV}_t)$, derive the attenuation of the OLS slope.
  2. Illustration (signal variance $\sigma^2_{V^*}$ assumed not to depend on $\mathrm{IQ}_t$): derive the best linear weight on $\mathrm{RV}_t$ given $\mathrm{IQ}_t$ and show that it falls with $\mathrm{IQ}_t$.
  3. Write an instrumental-variables estimator with instrument $\mathrm{RV}_{t-\mathbf{1}}$ and state its exclusion and relevance conditions.
- **Report**: the attenuation formula and its limiting cases, explaining the role of measurement uncertainty

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# B6 [Proposed]: Do Jumps Help? The HAR-CJ Model — Task

- **Prerequisites and examples**: B2 and B3 [Solved]
- **Data**: SPY as in B3; replication baseline: B2's 0.1% jump days, without corrections; $J_t = \mathrm{RV}_t - \mathrm{BV}_t$ on those days, 0 otherwise; $C_t = \mathrm{RV}_t - J_t + r^{2}_{\mathbf{on},t}$ (all in $\%^{2}$) Andersen, Bollerslev & Diebold (2007)
- **Tasks**
  1. Regress $\ln V_{t+\mathbf{1}}$ on $\ln C_t$, $\ln C_t^{(\mathbf{5})}$, $\ln C_t^{(\mathbf{22})}$, $\ln(1+J_t)$, $\ln(1+J_t^{(\mathbf{5})})$, $\ln(1+J_t^{(\mathbf{22})})$, where (5), (22) are arithmetic means logged afterwards; report all coefficients with Newey–West SEs and test the three jump coefficients jointly.
  2. Forecast from 3 January 2022 on an expanding window, back-transform with $\exp(\hat{y} + \hat{s}^{2}/2)$ as in B3, and compare QLIKE with log-HAR on common dates by DM.
  3. Sensitivity: repeat with the FDR-controlled, periodicity-corrected jump days of B2 extended.
- **Interpretation**: HAR-CJ averages levels before logging, log-HAR averages logs: the two are not nested; why can a significant jump coefficient give no forecasting gain?
- **Report**: coefficient table, joint Wald test, forecast losses and DM comparison; state whether significance translates into forecasting gains

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# A5 Extended [Solved]: Why MSE on Volatility Is Not Robust — Task

- **Prerequisites and examples**: A5 [Solved]
- **Inputs**: constant IV and $\mathrm{RV} = \mathrm{IV}\,\chi^2_M/M$; supplied moment: $\mathbb{E}\sqrt{\chi^2_M} = \sqrt{2}\,\Gamma(\frac{M+1}{2})/\Gamma(\frac{M}{2})$, $\Gamma$: the gamma function
- **Tasks**
  1. Minimise $\mathbb{E}(\sqrt{\mathrm{RV}} - \sqrt{F})^2$ (MSE on volatility) over $F$ and compare the minimiser with $\mathbb{E}\mathrm{RV}$ by Jensen's inequality.
  2. Compute $F^*/\mathrm{IV}$ for $M = 1$ (squared daily return), 6 and 78.
- **Report**: the optimal $F^*$, its comparison with $\mathbb{E}\mathrm{RV}$ and the three ratios $F^*/\mathrm{IV}$

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## A5 Extended [Solved]: Solution — Why MSE on Volatility Is Not Robust

- $\mathbb{E}(\sqrt{\mathrm{RV}} - \sqrt{F})^2 = \mathbb{E}\mathrm{RV} - 2\sqrt{F}\,\mathbb{E}\sqrt{\mathrm{RV}} + F$, minimal at $\sqrt{F} = \mathbb{E}\sqrt{\mathrm{RV}}$; by Jensen $(\mathbb{E}\sqrt{\mathrm{RV}})^2 < \mathbb{E}\mathrm{RV}$: the optimal forecast is biased down
- $F^*/\mathrm{IV} = \frac{2}{M}\left(\Gamma(\frac{M+1}{2})/\Gamma(\frac{M}{2})\right)^2 = 0.637$ ($M = 1$), $0.920$ ($M = 6$), $0.994$ ($M = 78$); plotted in A5
- **Interpretation**: against squared daily returns, MSE on volatility rewards a forecast 36% too low; with 5-minute RV the distortion almost disappears

# B7 [Proposed]: Forecasting the Next Week — Task

- **Prerequisites and examples**: B3 [Solved], A6, the evaluation workflow
- **Data**: SPY total variance $V_t$ (B3) and SPY adjusted closes since 2000 for GARCH; target $Y_t = V_t + \ldots + V_{t+4}$, forecast at the close of $t-1$; 1 176 origins from 3 January 2022
- **Tasks**
  1. Direct log-HAR: regress $\ln Y$ on the B3 regressors, re-estimated at every origin using only targets fully observed by $t-1$ (the last four are not); forecast $\exp(\hat{y} + \hat{s}^2/2)$ (lognormal correction).
  2. GARCH(1,1)-$t$ re-estimated every 21 days: with $\sigma_t^2$, $\omega$, $a$, $b$ as in Out-of-sample forecasts and the long-run variance $\bar{\sigma}^2 = \omega/(1-a-b)$, forecast $\sum_{h=0}^{4}[\bar{\sigma}^2 + (a+b)^h(\sigma_t^2 - \bar{\sigma}^2)]$.
  3. Compute QLIKE and MSE, DM on QLIKE with HAC errors, both MZ calibration coefficients (here $a$ is the intercept and $b$ the slope, distinct from the GARCH coefficients in step 2) with the joint test of $(a, b) = (0, 1)$, and plot the cumulative QLIKE difference.
- **Interpretation**
  - does the one-day advantage of log-HAR survive at a weekly horizon?
  - what do overlapping targets do to the DM test?
- **Report**: QLIKE/MSE table; DM statistic and p-value; MZ coefficients and joint p-value; cumulative QLIKE-difference chart and two interpretation sentences

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## From Conditional Losses to a Joint Test

- At one origin, instruments $Z = (1, 0.2)$ and loss difference $d = -0.1$ give $g = Zd = (-0.1, -0.02)$
- The second component tests whether past differences predict future ones
- Teaching example: $n = 100$, $\bar{g} = (0.02, -0.01)$, $\hat{\Gamma}_0 = \mathrm{diag}(0.02, 0.006)$, $\hat{\Gamma}_1 = \mathrm{diag}(0.01, 0.002)$, $\hat{\Gamma}_2 = \hat{\Gamma}_3 = \hat{\Gamma}_4 = 0$ ($\hat{\Gamma}_k$: the autocovariance of $g$ at lag $k$)
- $\hat{\Omega} = \hat{\Gamma}_0 + \sum_{k=1}^{4}(\hat{\Gamma}_k + \hat{\Gamma}_k^\top) = \mathrm{diag}(0.04, 0.01)$, so $W = 100[0.02^2/0.04 + (-0.01)^2/0.01] = 2$
- Against $\chi^2_2$, $p = e^{-1} \approx 0.368$: no rejection at 5%
- A non-rejection does not show that the methods are equally accurate in every state

# B7 Extended [Proposed]: A Conditional Giacomini–White Test — Task

- **Prerequisites and examples**: the numerical example immediately before this task; B7 supplies the forecasting setup
- Use the same sign as A6: QLIKE(log-HAR) minus QLIKE(GARCH). Under equal conditional predictive ability, compare the statistic with $\chi^2_2$; report its p-value and the 5% decision. A positive conditional mean loss difference favours GARCH.
- **Tasks**
  1. Re-estimate both methods on fixed-length rolling windows (log-HAR: last 250 targets; GARCH: last 1 000 returns), as the theory of Giacomini & White (2006) requires.
  2. With instruments $Z_t = (1, d_{t-5})$, known at origin $t-1$, and the loss difference $d_t$ of the target $Y_t$, known after the close of $t+4$, compute $n\,\bar{g}^\top \hat{\Omega}^{-1} \bar{g}$ with $g_t = Z_t d_t$ ($n$: number of origins, $\bar{g}$: the mean of $g$, $\hat{\Omega}$: its long-run covariance); under the null $g$ is uncorrelated beyond lag 4, so $\hat{\Omega}$ sums the unweighted autocovariances of lags 0–4.
- **Report**: the mean QLIKE losses, the Giacomini–White statistic, its p-value, the 5% decision and the unconditional test

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## Comparable Forecasts before the MCS Test

- Model Confidence Set (MCS): keeps the forecasts that the joint comparison cannot separate
- SHAR is estimated on variance levels, on an expanding window, with the first forecast on 3 January 2022
- The intercept and the weekly and monthly terms are as in HAR; the positive and negative semivariances replace the daily term
- Use only outcomes observed before the forecast date
- A forecast outside the $[\min, \max]$ range of the estimation-sample variance is replaced by the sample mean, as for HARQ
- `arch.bootstrap.MCS`, with the stationary bootstrap (mean block length 20): resample whole rows, jointly for all models
- Report the elimination order and both sets, 90% and 75%; explain why a model can stay in the first but not in the second

# B8 [Proposed]: Which Models Survive? The Model Confidence Set — Task

- **Prerequisites and examples**: B3 extended, B7
- **Data**: SPY one-day forecasts (log-HAR, HAR, GARCH(1,1)-$t$, EWMA, yesterday's RV) plus HARQ (B3 extended) and SHAR Patton & Sheppard (2015): $\beta_d V_t$ replaced by $\beta^{+}\mathrm{RS}_t^{+} + \beta^{-}\mathrm{RS}_t^{-}$, $\mathrm{RS}_t^{\pm} = \sum_i r_{t,i}^{2} 1\{r_{t,i} \gtrless 0\}$, overnight return counted as one more return; 1 180 common dates
- **Tasks**
  1. Build the date $\times$ model QLIKE matrix on common dates and check that every forecast is positive.
  2. Run the MCS Hansen, Lunde & Nason (2011) with the range statistic, 20-day blocks, 10 000 resamples, seed 42: `MCS(L, size=0.10, reps=10000, block_size=20, method='R', seed=42)`.
  3. Report the elimination order and every model's MCS $p$-value; compare the sets at 90% (size 0.10) and 75% (size 0.25).
- **Interpretation**: why is the MCS preferable to 21 pairwise DM tests?

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# B9 [Proposed]: How Rough Is Volatility? — Task

- **Prerequisites and examples**: B1 [Solved] (block bootstrap), Chapter 9
- **Data**: SPY daily series of four measures: total variance, intraday RV, BV, 30-minute subsampled RV; $x_t = \frac{1}{2}\ln(\text{measure}_t)$; lags $\Delta = 1, \ldots, 30$ trading days Gatheral, Jaisson & Rosenbaum (2018)
- **Tasks**
  1. Compute $m(q, \Delta)$, the mean of $|x_{t+\Delta} - x_t|^q$ over all valid pairs, for $q \in \{0.5, 1, 1.5, 2, 3\}$; check $m(2, 1)$ for total variance against the direct mean of squared first differences, using the same valid pairs.
  2. For each $q$ regress $\ln m(q, \Delta)$ on $\ln \Delta$ with an intercept: slope $\zeta_q$; then $H$ = slope of $\zeta_q$ on $q$ through the origin.
  3. 95% CI: moving-block bootstrap, 60-day blocks, 300 resamples, increments computed only within each block.
  4. Simulate a path as long as the SPY total-variance series, using `numpy.random.default_rng(42)` and lags 1–30. Use Gaussian random-walk increments ($N(0, 0.1^2)$, seed 42), subtract `linspace(0,z[-1],n)` from the cumulative path $z$ (still $H = 0.5$), estimate $H$, then draw i.i.d. $N(0, 0.25^2)$ noise from the same generator after the increments. Add it to the detrended path without detrending again; estimate $H$ again.
- **Interpretation**
  - is $H$ below 0.5 for every measure?
  - what does measurement error do?
- One path is an illustration, not a size or power study
- **Report**: H estimates and confidence intervals for every measure, scaling plots, and clean/noisy simulation comparison

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# C2 [Proposed]: Is Bitcoin's Volatility More Predictable Than the S&P 500's? — Task

- **Data**: SPY and Bitcoin one-day forecasts, common calendar window 1 March 2025 – 18 September 2026 (567 calendar days: SPY 389 trading days, Bitcoin 422 days)
- **Tasks**
  1. Define three measures and compute them per asset: the calibration $R^2$ of $\ln V_t$ on $\ln F_t$; the QLIKE gain $1 - \overline{\text{QLIKE}}_{\textbf{log-HAR}} / \overline{\text{QLIKE}}_{\textbf{RV yest.}}$; the same gain over GARCH.
  2. Build a 95% CI for the $R^2$ difference (Bitcoin minus SPY): resample the same 28-calendar-day blocks for both assets (2 000 times), keeping each asset's own missing days.
  3. Add a dummy for a weekend target day to the Bitcoin log-HAR, use 7/30-day HAR averages and an expanding training window of all complete observations before each target date, starting on 1 March 2025; forecast again with the same lognormal correction and run DM under QLIKE with and without the dummy.
- **Report**: the table, the CI, the DM result and a two-sentence answer; project extension: Ethereum or a Bursa de Valori București (BVB) stock

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# C3 [Proposed]: Does Machine Learning Beat HAR for SPY? — Project Brief (1/4)

- Context: Christensen, Siggaard & Veliyev (2023) (Chapter 9, case study)
  - random forests (RF) and neural networks beat HAR once predictors beyond past RV enter
  - the gain grows with the horizon
  - you test this on SPY, with the sample split and the model settings of the paper
- (a) Build the target series (the paper's Data description section).
  - daily open-to-close RV of SPY from the 5-minute bars of the regular session (78 returns a day, as in B2)
  - three targets: $\mathrm{RV}_{t+1}$, the average RV over the next 5 days and over the next 22 days
- (b) Build two information sets dated $t$.
  - $\mathcal{M}_{\mathsf{HAR}}$: the daily, weekly and monthly RV
  - $\mathcal{M}_{\mathsf{ALL}}$ adds the VIX (VIX.INDX.csv, close), the squared daily log return of the Hang Seng (HSI.INDX.csv, close) and the 1-week SPY momentum
  - and the first differences of log dollar volume and of the 3-month T-bill rate (Federal Reserve Economic Data, FRED; series `DTB3`)
  - and the economic policy uncertainty index (FRED, `USEPUINXD`) and the Aruoba–Diebold–Scotti index (Philadelphia Fed)

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# C3 [Proposed]: Does Machine Learning Beat HAR for SPY? — Project Brief (2/4)

- (b, continued) Date each predictor by its release, not its reference date.
  - use only values published by the close of day $t$
  - the Aruoba–Diebold–Scotti index is revised: use the Philadelphia Fed's real-time vintages
  - disclose any series for which vintages are unavailable
  - state which predictors of the paper (Table 1) cannot be built for an index fund, and why: a data limitation, not a reason to change the method
- (c) Split the sample as in Appendix A.1 of the paper.
  - the first 1 000 usable days for training, the next 200 for validation, the rest for the test
  - a row belongs to a subset only if its whole target window lies inside it: drop the last $h - 1$ rows before each boundary
  - report the usable sample size per horizon
  - standardise every predictor with the training mean and variance only

# C3 [Proposed]: Does Machine Learning Beat HAR for SPY? — Project Brief (3/4)

- (d) Estimate the linear models.
  - HAR, log-HAR and OLS on all predictors, on a rolling window over training and validation merged
  - log-HAR forecast: $\exp(\hat{f} + \hat{\sigma}^2/2)$, $\hat{f}$: fitted log RV, $\hat{\sigma}^2$: residual variance
  - elastic net on a rolling training window, without merging; $\lambda$ and the mixing weight chosen on the validation set (paper, Section 2)
  - HAR keeps its three RV regressors under both information sets
- (d, continued) Estimate the machine-learning models (paper, Table A.6).
  - random forest: 500 trees on the merged window, $\lfloor J/3 \rfloor$ candidate predictors per split ($J$: number of predictors), minimum node size 5
  - $\mathrm{NN}_2^{10}$: layers of 4 and 2 neurons, leaky ReLU, Adam with learning rate 0.001, dropout 0.8 (probability of removing a neuron during training; all neurons active at prediction)
  - early stopping with patience 100; the average of the best 10 of 100 seeds; trained once on a fixed window
  - replace negative forecasts with the smallest in-sample RV

## C3 [Proposed]: Does Machine Learning Beat HAR for SPY? — Project Brief (4/4)

- (e) Report the test-set comparison, as in B8.
  - a table of MSE and QLIKE relative to HAR, for the three horizons and both information sets
  - one-sided DM $p$-values against HAR and the 90% MCS
  - overlapping 5- and 22-day targets: HAC errors with a bandwidth above $h-1$ (Bartlett weights down-weight lag $h-1$) and a sensitivity check over bandwidths
- Interpretation
  - does the ranking of Tables 2, 3 and 7 of the paper survive on SPY?
  - does the ML advantage grow with the horizon?
  - does the ML advantage survive QLIKE?
  - with a test set of this length, what can a non-significant DM test tell you?
- Worked example to follow: B3, A5.
- Project extension
  - repeat on Bitcoin's 24-hour RV
  - compare across models the importance of RV and VIX from accumulated local effects (Figure 7 of the paper)

# Acronyms used in this material (1/2)

- **ACF**: AutoCorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **BH**: Benjamini–Hochberg (procedure)
- **BV**: Bipower Variation
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CI**: Confidence Interval
- **DM**: Diebold–Mariano (test)
- **ETF**: Exchange-Traded Fund
- **EWMA**: Exponentially Weighted Moving Average
- **FDR**: False Discovery Rate
- **FRED**: Federal Reserve Economic Data
- **FWER**: Family-Wise Error Rate
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HAR-CJ**: HAR model with Continuous and Jump components
- **HARQ**: HAR model with Realised Quarticity

# Acronyms used in this material (2/2)

- **IQ**: Integrated Quarticity
- **JB**: Jarque–Bera (test)
- **MCS**: Model Confidence Set
- **ML**: Machine Learning
- **MSE**: Mean Squared Error
- **MZ**: Mincer–Zarnowitz (regression)
- **NW**: Newey–West (standard errors)
- **OLS**: Ordinary Least Squares
- **QLIKE**: Quasi-LIKElihood (loss function)
- **QQ**: Quantile–Quantile (plot)
- **RF**: Random Forest
- **RQ**: Realised Quarticity
- **RV**: Realised Variance
- **SE**: Standard Error
- **SHAR**: Semivariance HAR model
- **TQ**: Tripower Quarticity
- **TSRV**: Two-Scale Realised Variance
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VIX**: CBOE Volatility IndeX

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